Target observability: technical derivation

This note is a software-oriented technical supplement to Stochastic Histories at Finite Temporal Resolution: Homological Observability and Information Phase Diagrams in Markov Jump Processes. The exposure filtration and hidden-history entropy law come from the paper. The target-image, conditional-entropy and minimum-readout arguments below are derived here for target diagnosis, readout design and scientific ML applications.

1. Full cycle group, sufficiency and certificates

Let a finite connected simple undirected support graph have a positive transition rate in both directions along each edge. Fix one orientation of each edge and a spanning tree. Its fundamental-cycle columns B identify the integer cycle group H with $\mathbb{Z}^b$, where $b = |E| - |V| + 1$. An integer cochain integrates signed edge counts; write $P : H \to \mathbb{Z}^k$ for fixed readouts and $F : H \to \mathbb{Z}^q$ for the targets.

All targets are determined by P on the full cycle group exactly when

$$\ker_{\mathbb{Z}} P \subseteq \ker_{\mathbb{Z}} F \quad \Longleftrightarrow \quad \operatorname{rank}_{\mathbb{Q}} P = \operatorname{rank}_{\mathbb{Q}} \begin{bmatrix} P \\ F \end{bmatrix}. \tag{1}$$

For the forward direction, every rational vector in $\ker P$ becomes an integer kernel vector after multiplication by a common denominator. Thus F annihilates the rational kernel, so its rows lie in P's rational row span. Conversely, that row-span condition gives a rational matrix D with $DP = F$; $Pz = 0$ implies $Fz = 0$. The code verifies $DP = F$ exactly before returning the decoder. If ranks differ, a saturated integer kernel basis has a column z with $Fz \neq 0$. Returning z, Bz and Fz witnesses the insufficiency; the product verifies $Pz = 0$ and $Fz \neq 0$.

`analyze_target` applies this criterion uniformly to **all** records and trajectories on the stated support. It works directly with fundamental-cycle coordinates. Additional restrictions from a particular finite observation enter its conditional history set. A spanning-tree change is an integral unimodular coordinate change, preserving the criterion and physical cochains.

2. Legal integer images and open trajectories

A rational decoder is defined on the legal integer image $P(H)$. The evaluator first solves $Pc = p$ over $\mathbb{Z}$ via a verified Smith decomposition $UPV = S$. Divisibility by each nonzero diagonal entry and vanishing of every residual row are necessary and sufficient for membership. It constructs and verifies an integer preimage. For example $P(x) = 2x$, $F(x) = x$ permits decoding $x = p/2$ for even p. With $P = 2I_2$, $F = (1, 1)$ and $p = (1, 1)$, the record lies outside $P(H) = 2\mathbb{Z}^2$, even though the decoder formula returns the integer 1.

For an open walk chain x from s to t, let $h(s, t)$ be the tree path with the same boundary. Then $x - h = Bc$ for a unique integer c. For raw cochain matrices A and G, the evaluator uses

$$p' = Ax - Ah = Pc, \qquad Gx = Dp' + Gh, \qquad P = AB,\ F = GB. \tag{2}$$

Known endpoints therefore determine pure-gradient targets, whose cycle restriction is zero. `record_kind="cycle"` accepts the corrected coordinates without endpoint arguments and returns the cycle contribution Fc.

3. The exact target ambiguity image

For a legal p choose $c_0 \in H$ with $Pc_0 = p$. The entire integer solution set is $c_0 + K$, where $K = \ker_{\mathbb{Z}} P$. Consequently the target set is precisely

$$F(c_0) + M, \qquad M = F(K). \tag{3}$$

Both inclusions follow directly by subtracting c_0 or adding a kernel vector. The choice of c_0 changes the representative but leaves the coset unchanged. Open-path targets add the known tree-path term. The kernel must be saturated: independently clearing denominators in rational-nullspace columns can generate only a proper sublattice. For $P = (2, 1, 1)$, $F = (0, 1, 0)$, the true image M is $\mathbb{Z}$; a naive basis can incorrectly return $2\mathbb{Z}$. The Smith-based kernel and Hermite image backend preserve this information.

The discrete flashing ratchet (DFR) example has two triangular rings (0,1,2) and (3,4,5), with switching edges $i \leftrightarrow i + 3$. The counters W_{ON} and W_{OFF} count signed crossings of $2 \to 0$ and $5 \to 3$, respectively. With $W = W_{\mathrm{ON}} + W_{\mathrm{OFF}}$ and $\Delta W = W_{\mathrm{ON}} - W_{\mathrm{OFF}}$, we have $\Delta W \equiv W \pmod 2$. At fixed W the image is $2\mathbb{Z}$: a unit ON circulation and a reverse OFF circulation preserve W while changing ΔW by 2. These loops can be connected by opposite traversals of a switching edge, which cancel in every cochain. For endpoints $0 \to 3$ the walks [0,3] and [0,1,2,0,3,5,4,3,0,3] have $W = 0$ and ΔW equal to 0 and 2. The packaged pressure case completes all their transitions before the first positive snapshot time; every snapshot, including endpoints, agrees.

4. Target-image filtration and entropy

Use the paper's integer exposure filtration L_m. The existing backend constructs it from simple cycles of length at most $2m$. Define

$$K_P^m = L_m \cap \ker P, \qquad K_{P,F}^m = L_m \cap \ker P \cap \ker F, \qquad M_m = F(K_P^m).$$

Restriction of F gives a short exact sequence

$$0 \longrightarrow K_{P,F}^m \longrightarrow K_P^m \longrightarrow M_m \longrightarrow 0.$$

Ranks therefore add at each m. Nested L_m imply nested K_P^m and M_m, so the graded target rank

$$g_m = \operatorname{rank} M_m - \operatorname{rank} M_{m-1} = r_P^m - r_{P,F}^m \geq 0$$

is nonnegative. Here r denotes the corresponding hidden graded free rank. The implementation constructs the integer images, verifies lattice inclusion, and checks this rank identity. The free ranks determine the $\log T$ leading coefficient; the integer lattices also retain finite-index information such as parity.

Let C be the latent integer cycle history, S the snapshots and PC the observed cochain record. Since FC is a function of C, the conditional entropy chain rule is

$$H(FC \mid S, PC) = H(C \mid S, PC) - H(C \mid S, PC, FC) = R_P - R_{(P,F)}.$$

Under the original paper's fixed finite connected bidirectional positive-rate model, stationary initial law and regular noiseless snapshots, Theorem 5.1 applied to both entropies on the **same** model gives, for fixed $\alpha > 0$, $\delta = T^{-\alpha}$ and $T/\delta \in \mathbb{N}$ as $T \to \infty$,

$$H(FC \mid S, PC) = \frac{1}{2} \sum_{m \geq 2} g_m [1 - \alpha(m-1)]_+ \log T + o(\log T). \tag{4}$$

For a fixed snapshot interval $\delta_0 > 0$, Proposition 5.3 applied to the same two entropies instead gives, along $T/\delta_0 \in \mathbb{N}$,

$$H(FC \mid S, PC) = \frac{1}{2} \operatorname{rank} F(\ker P) \log T + O(1).$$

Here $\operatorname{rank} F(\ker P) = \sum_m g_m$, matching the leading coefficient at $\alpha = 0$. At a layer carrying target rank, the critical exponent is $\alpha = 1/(m-1)$. A zero coefficient gives sublogarithmic target entropy in this limit. Its residual uncertainty calls for finer asymptotic analysis, including the conditional limit law at critical scales. For finite inputs T, δ, the reported $T\delta^{m-1}$ measures exposure at each layer. Recovery probabilities depend on the dynamics and the conditional history law; exact reconstruction from counters follows the criterion in section 1.

The global coefficient is available once the computed exposure lattice reaches H; an earlier stop returns a prefix coefficient and `PARTIAL_HORIZON`. The default horizon is $\lceil |V|/2 \rceil$, but early saturation also suffices. Explanation budgets cover cycles, matrices, integer sizes and cooperative time; explanation status is independent of the core certificate. A design explanation uses the **fixed** readouts. To explain a selected augmentation, re-run analysis with its added channels.

5. Minimum additional scalar channels

The existing readout P_0 is fixed. Candidate rows have unique physical IDs; distinct instruments with equal cochains remain distinct choices. For each subset J test sufficiency of $[P_0; P_J]$. If P_0 already suffices, the empty addition is optimal. If the entire pool fails, no subset can succeed by monotonicity. Otherwise search cardinalities in increasing order. Exhausting all smaller sizes before the first success proves global cardinality minimality. Exhausting the successful size establishes all ties. Constructing each returned certificate is a separate operation.

Thus feasibility, minimum proof, tied-set enumeration, output truncation and certificate completion have independent fields. Stopping mid-size retains established lower/upper bounds and any proven minimum. A count of discovered ties equals the total after that size's enumeration is complete. The objective is the number of added scalar channels, with each candidate counted once. When all events are available, the candidate pool can include a direct target counter. Fixed records, restricted instruments and shared targets determine which choices are available in a particular application.

6. Equal-input prediction reference

On the constructed equal-weight pair with targets 0 and 2, a deterministic function of the identical visible input returns a common real a. Its mean absolute error obeys $(|a|+|2-a|)/2 \geq 1$ by the triangle inequality. It can equal at most one target, giving exact-match accuracy $\leq 1/2$. For randomized predictors whose randomness is independent of the hidden label, these bounds hold after averaging over prediction randomness as well as the equal-weight pair.

The pair's conditional mean is 1, while its realized targets are 0 and 2. The evaluation adapter distinguishes conditional means, real point predictions and realized integer values, preserves original values, and scores explicitly rounded predictions separately. The bounds above use the constructed pair's equal weights. Natural CTMC predictions are evaluated on their own sampled trajectories and reported separately.

7. Executable evidence

- `tests/test_target_tasks.py`:
  independent Fraction elimination, exhaustive DFR subsets, saturated-kernel and illegal-image counterexamples, integer witness checks, finite open-walk reconstruction and alternate trees, and search bounds.
- `tests/test_target_explanation.py`:
  target-image rank monotonicity and agreement with the difference of two existing hidden-history analyses across DFR and the $C_3 \vee C_5$ wedge, exact α boundaries, alternate trees, pure-gradient targets, limited horizons, independent core/explanation status, schemas and replay.
- `tests/test_dfr_application.py`:
  support and stationary law for a known generator Q, direct simulation against an independent two-state matrix exponential, reproducible seeds, whole-trajectory pairing, full-grid ambiguity, feature separation, train-only baseline fitting, tampering/identity failures, point semantics, exact reconstruction, prediction replay and CLI behavior.

Run `python -W error -m unittest discover -s tests -v` after source installation. The application tutorial reproduces the fixed-data measurements.